

COMPARATIVE ANALYSIS

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Course : Computer Vision

1. Object Detection Performance Comparison

Given Data

Two object detection models, Model A and Model B, have the following results:

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
Model A	80	20	30
Model B	90	15	20

The task requires calculating precision and recall and comparing the overall detection performance.

(a) Precision and Recall

The formulas are:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Model A

Precision:

$$\text{Precision} = 80 / (80 + 20)$$

$$= 80 / 100$$

$$= 0.80 = 80\%$$

Recall:

$$\text{Recall} = 80 / (80 + 30)$$

$$= 80 / 110$$

$$= 0.7273 = 72.73\%$$

Model B**Precision:**

$$\text{Precision} = 90 / (90 + 15)$$

$$= 90 / 105$$

$$= 0.8571 = 85.71\%$$

Recall:

$$\text{Recall} = 90 / (90 + 20)$$

$$= 90 / 110$$

$$= 0.8182 = 81.82\%$$

(b) Comparison

Metric	Model A	Model B
Precision	80%	85.71%
Recall	72.73%	81.82%

Model B has higher precision and higher recall than Model A.

(c) Justification

Model B provides better overall object detection performance because:

- Its precision is higher, meaning it produces fewer incorrect positive detections.
- Its recall is higher, meaning it detects a larger proportion of the actual objects.
- Model B has more True Positives and fewer False Positives and False Negatives compared with Model A.

Conclusion

Model B is the better object detection model because it achieves both higher precision and higher recall. Therefore, it provides more reliable overall object detection performance.

2. Motion Estimation Error Analysis

Given Data

The displacement errors for three consecutive frames are:

- Method A: 2, 3, 2 pixels
- Method B: 3, 5, 4 pixels

The task requires calculating the average displacement error and determining which method provides more accurate motion estimation.

(a) Average Displacement Error

The formula is:

Average Error = Sum of Errors / Number of Frames

Method A

Average Error = $(2 + 3 + 2) / 3$

= $7 / 3$

= **2.33 pixels**

Method B

Average Error = $(3 + 5 + 4) / 3$

= $12 / 3$

= **4 pixels**

(b) Comparison

Method	Errors	Average Error
Method A	2, 3, 2 pixels	2.33 pixels
Method B	3, 5, 4 pixels	4 pixels

Method A has a lower average displacement error than Method B.

(c) Justification

A lower displacement error indicates that the estimated motion is closer to the actual motion. Method A has an average error of only 2.33 pixels, while Method B has an average error of 4 pixels.

Therefore, Method A provides more accurate motion estimation.

Conclusion

Method A is the better motion estimation technique because it has a lower average displacement error. Its errors are also consistently smaller across all three frames.

3. Depth Estimation Comparison

Given Data

The stereo vision system and Structure-from-Motion (SfM) system estimate the depth of three objects.

Object	Actual Depth	Stereo Vision	SfM
1	2.0 m	2.0 m	2.5 m
2	3.0 m	3.0 m	3.8 m
3	4.0 m	4.0 m	5.0 m

(a) Absolute Error

The formula is:

$$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$$

Object 1

Stereo Vision:

$$|2.0 - 2.0| = 0.0 \text{ m}$$

SfM:

$$|2.5 - 2.0| = 0.5 \text{ m}$$

Object 2

Stereo Vision:

$$|3.0 - 3.0| = 0.0 \text{ m}$$

SfM:

$$|3.8 - 3.0| = 0.8 \text{ m}$$

Object 3

Stereo Vision:

$$|4.0 - 4.0| = 0.0 \text{ m}$$

SfM:

$$|5.0 - 4.0| = 1.0 \text{ m}$$

Error Comparison

Object	Stereo Vision Error	SfM Error
1	0.0 m	0.5 m
2	0.0 m	0.8 m
3	0.0 m	1.0 m

(b) Average Absolute Error

Stereo Vision

$$\text{Average Error} = (0.0 + 0.0 + 0.0) / 3$$

$$= 0.0 \text{ m}$$

SfM

$$\text{Average Error} = (0.5 + 0.8 + 1.0) / 3$$

$$= 2.3 / 3$$

$$= 0.767 \text{ m}$$

(c) Comparison

Method	Average Absolute Error
Stereo Vision	0.0 m
SfM	0.767 m

Stereo Vision has zero error for all three objects, whereas SfM has errors ranging from 0.5 m to 1.0 m.

Conclusion

Stereo Vision provides more accurate depth estimation because its average absolute error is 0.0 m, compared with approximately 0.767 m for SfM. Therefore, based on the given data, Stereo Vision is the more accurate method.